

Hands-on Ingestion Tasks

```
1 #starter code
2 from pyspark.sql import SparkSession
3
4 spark = SparkSession.builder \
5     .appName("Phase2") \
6     .getOrCreate()
7
8 # 1. Read a CSV file from /samples/
9 df = spark.read.format("csv") \
10     .option("header", "true") \
11     .load("/samples/customers.csv")
12 df.show()
13
14
```

customer_id	first_name	last_name	email	phone_number	address	city	state	zip_code
1	John	Smith	john.smith@domain...	555-0001	123 Elm St	Springfield	IL	62701
2	Emma	Jones	emma.jones@webmai...	555-0002	456 Oak St	Centerville	OH	45459
3	Olivia	Brown	olivia.brown@outl...	555-0003	789 Pine St	Greenville	SC	29601
4	Liam	Johnson	liam.johnson@gmai...	555-0004	101 Maple St	Riverside	CA	92501
5	Noah	Williams	noah.williams@yah...	555-0005	202 Birch St	Lakeside	TX	75001
6	Alice	Miller	alice.miller@aol.com	555-0006	303 Cedar St	Oakland	CA	94601
7	Isabella	Davis	isabella.davis@ic...	555-0007	404 Spruce St	Boise	ID	83701
8	James	Martinez	james.martinez@li...	555-0008	505 Walnut St	Des Moines	IA	50301
9	Sophia	Garcia	sophia.garcia@zoh...	555-0009	606 Cherry St	Albany	NY	12201
10	Lucas	Rodriguez	lucas.rodriguez@h...	555-0010	707 Maple St	Portland	OR	97201
11	Mia	Lopez	mia.lopez@mail.com	555-0011	808 Oak St	Miami	FL	33101
12	William	Anderson	william.anderson@...	555-0012	909 Pine St	Nashville	TN	37201
13	Amelia	Thomas	amelia.thomas@pro...	555-0013	1010 Elm St	Denver	CO	80201
14	Ethan	Taylor	ethan.taylor@inbo...	555-0014	1111 Birch St	Minneapolis	MN	55401
15	Harper	Jackson	harper.jackson@ou...	555-0015	1212 Cedar St	Seattle	WA	98101
16	Jackson	White	jackson.white@yma...	555-0016	1313 Spruce St	Atlanta	GA	30301
17	Charlotte	Harris	charlotte.harris@...	555-0017	1414 Walnut St	San Diego	CA	92101
18	Oliver	Martin	oliver.martin@icl...	555-0018	1515 Cherry St	Indianapolis	IN	46201
19	Madison	Thompson	madison.thompson@...	555-0019	1616 Maple St	Charlotte	NC	28201
20	Elijah	Garcia	elijah.garcia@zoh...	555-0020	1717 Oak St	Detroit	MI	48201

only showing top 20 rows

```
#starter code
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("Phase3") \
    .getOrCreate()

#2. Inspect schema using show() and printSchema()
df = spark.read.format("csv") \
    .option("header", "true") \
    .load("/samples/customers.csv")
df.show()
df.printSchema()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
|customer_id|first_name|last_name|email|phone_number|address|city|state|zip_code|
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1| John| Smith|john.smith@domain...| 555-0001| 123 Elm St| Springfield| IL| 62701|
| 2| Emma| Jones|emma.jones@webmai...| 555-0002| 456 Oak St| Centerville| OH| 45459|
| 3| Olivia| Brown|olivia.brown@outl...| 555-0003| 789 Pine St| Greenville| SC| 29601|
| 4| Liam| Johnson|liam.johnson@gmai...| 555-0004| 101 Maple St| Riverside| CA| 92501|
| 5| Noah| Williams|noah.williams@yah...| 555-0005| 202 Birch St| Lakeside| TX| 75001|
| 6| Alice| Miller|alice.miller@aol.com| 555-0006| 303 Cedar St| Oakland| CA| 94601|
| 7| Isabella| Davis|isabella.davis@ic...| 555-0007| 404 Spruce St| Boise| ID| 83701|
| 8| James| Martinez|james.martinez@li...| 555-0008| 505 Walnut St| Des Moines| IA| 50301|
| 9| Sophia| Garcia|sophia.garcia@zoh...| 555-0009| 606 Cherry St| Albany| NY| 12201|
| 10| Lucas| Rodriguez|lucas.rodriguez@h...| 555-0010| 707 Maple St| Portland| OR| 97201|
| 11| Mia| Lopez| mia.lopez@mail.com| 555-0011| 808 Oak St| Miami| FL| 33101|
| 12| William| Anderson|william.anderson@...| 555-0012| 909 Pine St| Nashville| TN| 37201|
| 13| Amelia| Thomas|amelia.thomas@pro...| 555-0013| 1010 Elm St| Denver| CO| 80201|
| 14| Ethan| Taylor|ethan.taylor@inbo...| 555-0014| 1111 Birch St| Minneapolis| MN| 55401|
| 15| Harper| Jackson|harper.jackson@ou...| 555-0015| 1212 Cedar St| Seattle| WA| 98101|
| 16| Jackson| White|jackson.white@yma...| 555-0016| 1313 Spruce St| Atlanta| GA| 30301|
| 17| Charlotte| Harris|charlotte.harris@...| 555-0017| 1414 Walnut St| San Diego| CA| 92101|
| 18| Oliver| Martin|oliver.martin@icl...| 555-0018| 1515 Cherry St| Indianapolis| IN| 46201|
| 19| Madison| Thompson|madison.thompson@...| 555-0019| 1616 Maple St| Charlotte| NC| 28201|
| 20| Elijah| Garcia|elijah.garcia@zoh...| 555-0020| 1717 Oak St| Detroit| MI| 48201|
+-----+-----+-----+-----+-----+-----+-----+-----+
```

only showing top 20 rows

```
root
|-- customer_id: string (nullable = true)
|-- first_name: string (nullable = true)
|-- last_name: string (nullable = true)
|-- email: string (nullable = true)
|-- phone_number: string (nullable = true)
|-- address: string (nullable = true)
|-- city: string (nullable = true)
|-- state: string (nullable = true)
|-- zip_code: string (nullable = true)
```

```
#starter code
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("Phase3") \
    .getOrCreate()

df = spark.read.format("csv") \
    .option("header", "true") \
    .load("/samples/customers.csv")

#)4. Clean data using dropna() or fillna()
clean_df = df.dropna()
clean_df.show()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
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|1|John|Smith|john.smith@domain...|555-0001|123 Elm St|Springfield|IL|62701|
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|3|Olivia|Brown|olivia.brown@outl...|555-0003|789 Pine St|Greenville|SC|29601|
|4|Liam|Johnson|liam.johnson@gmai...|555-0004|101 Maple St|Riverside|CA|92501|
|5|Noah|Williams|noah.williams@yah...|555-0005|202 Birch St|Lakeside|TX|75001|
|6|Alice|Miller|alice.miller@aol.com|555-0006|303 Cedar St|Oakland|CA|94601|
|7|Isabella|Davis|isabella.davis@ic...|555-0007|404 Spruce St|Boise|ID|83701|
|8|James|Martinez|james.martinez@li...|555-0008|505 Walnut St|Des Moines|IA|50301|
|9|Sophia|Garcia|sophia.garcia@zoh...|555-0009|606 Cherry St|Albany|NY|12201|
|10|Lucas|Rodriguez|lucas.rodriguez@h...|555-0010|707 Maple St|Portland|OR|97201|
|11|Mia|Lopez|mia.lopez@mail.com|555-0011|808 Oak St|Miami|FL|33101|
|12|William|Anderson|william.anderson@...|555-0012|909 Pine St|Nashville|TN|37201|
|13|Amelia|Thomas|amelia.thomas@pro...|555-0013|1010 Elm St|Denver|CO|80201|
|14|Ethan|Taylor|ethan.taylor@inbo...|555-0014|1111 Birch St|Minneapolis|MN|55401|
|15|Harper|Jackson|harper.jackson@ou...|555-0015|1212 Cedar St|Seattle|WA|98101|
|16|Jackson|White|jackson.white@yma...|555-0016|1313 Spruce St|Atlanta|GA|30301|
|17|Charlotte|Harris|charlotte.harris@...|555-0017|1414 Walnut St|San Diego|CA|92101|
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+-----+-----+-----+-----+-----+-----+-----+-----+

only showing top 20 rows
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Business Pipeline Exercises

```
1  #starter code
2  from pyspark.sql import SparkSession
3
4  spark = SparkSession.builder \
5      .appName("Phase3") \
6      .getOrCreate()
7  #)1. Read sales data ->
8  sales=spark.read.option("header","true").csv("/samples/sales.csv")
9
10
11 # clean nulls ->
12 sales=sales.dropna()
13
14
15 #calculate daily sales
16 from pyspark.sql.functions import sum
17 sales=sales.groupBy("sale_date")\
18     .agg(sum("total_amount").alias("daily_sales"))
19 sales.show()
```

```
+-----+-----+
| sale_date|daily_sales|
+-----+-----+
| 2024-01-19|      66.75|
| 2024-02-08|      49.98|
| 2024-02-28|      67.47|
| 2024-02-20|      34.0|
| 2024-03-03|      55.5|
| 2024-02-15|      45.0|
| 2024-02-12|      40.0|
| 2024-02-26|      89.96|
| 2024-02-06|      29.99|
| 2024-01-17|      49.98|
| 2024-01-18|     119.96|
| 2024-02-16|      35.98|
| 2024-02-23|      18.75|
| 2024-01-16|      25.0|
| 2024-02-19|      27.5|
| 2024-01-29|      92.0|
| 2024-02-10|      15.5|
| 2024-02-21|      92.0|
| 2024-01-15|      39.98|
| 2024-03-01|      22.5|
+-----+-----+
```

only showing top 20 rows

```

1  #starter code
2  from pyspark.sql import SparkSession
3
4  spark = SparkSession.builder \
5      .appName("Phase3") \
6      .getOrCreate()
7
8  #1. Read customer data ->
9  cust=spark.read.option("header","true").csv("/samples/customers.csv")
10
11 #2.clean invalid rows ->
12 from pyspark.sql.functions import col,trim
13
14 #cleaning nulls
15 cust=cust.dropna()
16 #removing rows with empty values
17 cust=cust.filter(trim(col("first_name")) !="" )
18 # Remove duplicate customer IDs
19 cust=cust.dropDuplicates(["customer_id"])
20 cust.show()
21
22 #city-wise revenue
23 from pyspark.sql.functions import sum
24 sales=spark.read.option("header","true").csv("/samples/sales.csv")
25
26 city_revenue=cust.join(sales,how="inner", on="customer_id")\
27     .groupBy("city")\
28     .agg(sum("total_amount").alias("city_wise_revenue"))
29
30 city_revenue.show()

```

customer_id	first_name	last_name	email	phone_number	address	city	state	zip_code
1	John	Smith	john.smith@domain...	555-0001	123 Elm St	Springfield	IL	62701
10	Lucas	Rodriguez	lucas.rodriguez@h...	555-0010	707 Maple St	Portland	OR	97201
11	Mia	Lopez	mia.lopez@mail.com	555-0011	808 Oak St	Miami	FL	33101
12	William	Anderson	william.anderson@...	555-0012	909 Pine St	Nashville	TN	37201
13	Amelia	Thomas	amelia.thomas@pro...	555-0013	1010 Elm St	Denver	CO	80201
14	Ethan	Taylor	ethan.taylor@inbox...	555-0014	1111 Birch St	Minneapolis	MN	55401
15	Harper	Jackson	harper.jackson@ou...	555-0015	1212 Cedar St	Seattle	WA	98101
16	Jackson	White	jackson.white@yma...	555-0016	1313 Spruce St	Atlanta	GA	30301
17	Charlotte	Harris	charlotte.harris@...	555-0017	1414 Walnut St	San Diego	CA	92101
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19	Madison	Thompson	madison.thompson@...	555-0019	1616 Maple St	Charlotte	NC	28201
2	Emma	Jones	emma.jones@webmai...	555-0002	456 Oak St	Centerville	OH	45459
20	Elijah	Garcia	elijah.garcia@zoh...	555-0020	1717 Oak St	Detroit	MI	48201
21	Scarlett	Wilson	scarlett.wilson@l...	555-0021	1818 Pine St	Jacksonville	FL	32201
22	Henry	Moore	henry.moore@yahoo...	555-0022	1919 Elm St	Boston	MA	02101
23	Evelyn	Lee	evelyn.lee@aol.com	555-0023	2020 Birch St	Pittsburgh	PA	15201
24	Daniel	Walker	daniel.walker@fas...	555-0024	2121 Cedar St	St. Louis	MO	63101
25	Harper	Hall	harper.hall@inbox...	555-0025	2222 Spruce St	Honolulu	HI	96801
26	Matthew	Allen	matthew.allen@dom...	555-0026	2323 Walnut St	Milwaukee	WI	53201
27	Sofia	Young	sofia.young@webma...	555-0027	2424 Cherry St	San Antonio	TX	78201

only showing top 20 rows

-----+-----+	
city	city_wise_revenue
-----+-----+	
Springfield	69.97
Oakland	40.0
San Antonio	79.96
Philadelphia	60.0
Indianapolis	59.97
Nashville	67.47
San Diego	95.5
San Francisco	27.5
Detroit	89.97
Portland	55.0
Albany	79.96
Boise	110.95
St. Louis	99.95
Columbus	71.97
Memphis	89.96
Washington	92.0
El Paso	19.99
Milwaukee	66.75
Pittsburgh	15.5
Centerville	114.97
-----+-----+	

only showing top 20 rows

```

1  #starter code
2  from pyspark.sql import SparkSession
3
4  spark = SparkSession.builder \
5      .appName("Phase3") \
6      .getOrCreate()
7
8  #)3. Find repeat customers (>2 orders)
9  from pyspark.sql.functions import count, col
10
11 sales = spark.read.option("header", "true").csv("/samples/sales.csv")
12
13 rep_cust = sales.groupBy("customer_id") \
14     .agg(count("customer_id").alias("order_count")) \
15     .filter(col("order_count") > 2)
16
17 rep_cust.show()

```

```

+-----+-----+
|customer_id|order_count|
+-----+-----+
+-----+-----+

```

```
1  #starter code
2  from pyspark.sql import SparkSession
3
4  spark = SparkSession.builder \
5      .appName("Phase3") \
6      .getOrCreate()
7
8  #)4. Find highest spending customer in each city
9  from pyspark.sql.window import Window
10 from pyspark.sql.functions import sum,rank
11 customers = spark.read.option("header", "true").csv("/samples/customers.csv")
12 sales = spark.read.option("header", "true").csv("/samples/sales.csv")
13 cust=customers.join(sales,"customer_id")\
14     .groupBy("customer_id","city")\
15     .agg(sum("total_amount").alias("total_spent"))
16
17 window=Window.partitionBy("city").orderBy(cust.total_spent.desc())
18 result=cust.withColumn("rank",rank().over(window))\
19     .filter("rank=1")
20 result.show()
21
22 |
```

customer_id	city	total_spent	rank
9	Albany	79.96	1
16	Atlanta	60.0	1
36	Baltimore	18.75	1
7	Boise	110.95	1
22	Boston	119.96	1
2	Centerville	114.97	1
37	Charlotte	39.98	1
43	Chicago	40.0	1
30	Columbus	71.97	1
40	Denver	40.0	1
8	Des Moines	20.0	1
20	Detroit	89.97	1
38	El Paso	19.99	1
3	Greenville	49.98	1
25	Honolulu	40.0	1
42	Houston	22.5	1
18	Indianapolis	59.97	1
33	Jackson	34.0	1
21	Jacksonville	49.98	1
5	Lakeside	66.75	1

only showing top 20 rows

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import sum, count

spark = SparkSession.builder \
    .appName("Phase3") \
    .getOrCreate()
#)5. Build final reporting table with customer, city, total spend, order count
customers = spark.read.option("header", "true").csv("/samples/customers.csv")
sales = spark.read.option("header", "true").csv("/samples/sales.csv")

report = customers.join(sales, on="customer_id") \
    .groupBy("customer_id", "city") \
    .agg(
        sum("total_amount").alias("total_spend"),
        count("customer_id").alias("order_count")
    )

report.show()
```

customer_id	city	total_spend	order_count
5	Lakeside	66.75	1
27	San Antonio	79.96	1
4	Riverside	135.45999999999998	2
20	Detroit	89.97	1
44	San Diego	55.5	1
23	Pittsburgh	15.5	1
9	Albany	79.96	1
43	Chicago	40.0	1
11	Miami	25.0	1
39	Memphis	89.96	1
42	Houston	22.5	1
16	Atlanta	60.0	1
7	Boise	110.95	1
31	San Jose	50.0	1
37	Charlotte	39.98	1
1	Springfield	69.97	2
12	Nashville	67.47	1
30	Columbus	71.97	1
35	Philadelphia	60.0	1
15	Seattle	92.0	1

only showing top 20 rows